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Cisco Secure Workload — IEC 62443 Compliance Framework

Reference Design | Industrial Automation & Control Systems (IACS)

Version: 1.0

Use Case: Fresh Install, Hybrid Environment (On-Prem + Cloud)

Standards anchor: IEC 62443-3-3 (System security requirements and security levels); IEC 62443-2-1 (Security program requirements for IACS asset owners) — consult your SL-T target and zone/conduit documentation for authoritative control text.

Reader's Guide

Who this is for. IACS asset owners, OT/IT security architects, plant or operations cybersecurity teams, and integrators preparing for IEC 62443-aligned assessments, customer security requirements, or internal security management system (SMS) evidence for industrial environments.

Scope boundary you must understand before reading further. Cisco Secure Workload (CSW) enforces segmentation, discovers dependencies, and produces forensic telemetry on **servers, virtual machines, containers, and cloud workloads** — the **IT side of the IT/OT boundary** and the systems that **support** IACS (jump hosts, engineering workstations, historians, MES interfaces, DMZ brokers, identity services, patch servers, vendor remote-access concentrators). CSW does **not** replace dedicated OT visibility for Level 0–2 devices (PLCs, RTUs, IEDs, drives, field instruments). Pair CSW at the IT layer with **Cisco Cyber Vision, Claroty, Nozomi**, or equivalent for asset discovery, ICS protocol analytics, and device-level integrity signals.

Questions this reference design helps you answer:

- *SR 5.1–5.4 (restricted data flow / zones & conduits): Can I prove that only documented conduits carry traffic between OT-support tiers and corporate/cloud, and that lateral movement outside those conduits is structurally denied or logged?*
- *SR 1.1–1.13 (identification / authentication / access control): Can I show identity-aligned allow rules and deny-by-default paths for systems that administer or bridge into IACS zones?*
- *SR 3.1–3.9 (data / system integrity): Can I baseline inventory (processes, binaries, listening services) and detect unexpected change or unauthorized software on IACS-adjacent IT workloads?*

- *SR 6.1–6.2 (event monitoring & timely response): Can I reconstruct a timeline of flows and processes during an OT-adjacent incident?*
- *SR 7.1–7.8 (resource availability): Can I detect volumetric or connection storms indicative of denial-of-service against critical IT services that underpin OT availability?*
- *IEC 62443-2-1 (security program / operations): Can I produce continuous monitoring dashboards and exports that feed our SMS without manual spreadsheet reconciliation?*

What you'll need. Your zone & conduit model (per IEC 62443-3-2 or customer ZCRD), Security Level Target (SL-T) per zone, inventory of IACS-adjacent IT systems (EWS, jump hosts, AD/PKI, historians), change management approval for sensor install, and naming alignment between your PAS/OT tools and CMDB labels.

Where to start. Sections 1–2 while scoping; 3–5 during pilot sensor and baseline; 6–8 when designing enforcement; 9–10 for control mapping and SMS reporting; section **Boundaries** before promising CSW coverage to integrators or auditors.

CSW primer — if you are new to Cisco Secure Workload

Cisco Secure Workload (CSW) is a **workload protection platform**. A lightweight **agent** on each server/VM/container observes processes and network flows; **cloud connectors** add AWS/Azure/GCP inventory where agents are not deployed.

CSW term	Meaning	Why compliance teams care
Scope	Logical boundary (CDE, PHI zone, CUI enclave)	Defines the systems you must prove are isolated
Label / filter	Tag or query assigning workloads to scopes	Automates scope membership — reduces drift
ADM	Application Dependency Mapping from observed traffic	Live network/data-flow diagram for assessors
Workspace	Policy container for one scope or application	Where allow/deny rules are authored
Monitor → Simulation → Enforce	Safe rollout sequence	Simulation = change-board evidence before blocking traffic
Denied Connections	Log of flows blocked by policy	Primary proof that enforcement operates

Console areas: Investigate (inventory, flows, vulns) · Defend/Segmentation (policy) · Manage (agents) · Platform (connectors) · Administration (audit log)

Read next: [Compliance evidence playbook](#) (full step-by-step) · [About CSW](#) (platform intro)

Universal evidence workflow

Execute these phases for **this framework's** compliance boundary. Map exports to control IDs in the **Reporting & Evidence** section below.

Phase	Goal	Key CSW actions	Typical duration
1 — Coverage	Every in-scope workload in CSW	Install agents/connectors ; apply compliance labels; create scope	Days 1–10
2 — Baseline	Machine-generated flow map	Run ADM ≥ 2 weeks; export clusters; app-owner signoff	Days 11–28
3 — Policy	Designed isolation before enforce	Build workspace; Simulation mode; fix false positives	Days 29–45
4 — Operate	Continuous proof between audits	Enforce; quarterly export pack; ADM refresh every 90 days	Ongoing

Phase 1 checklist (coverage)

- ☐ In-scope host list reconciled to CSW Inventory (100% or documented exceptions)
- ☐ Labels applied: `compliance:<framework>`, `data:<classification>`, `env:<tier>`
- ☐ CSW scope created matching compliance boundary
- ☐ **Export:** inventory CSV + agent status screenshot

Phase 2 checklist (baseline)

- ☐ ADM running on compliance scope for full business cycle (≥ 2 weeks)
- ☐ Unexpected flows documented (shadow IT, vendor egress, scope creep)
- ☐ App owners signed cluster-to-application mapping
- ☐ **Export:** ADM diagram + flow samples with process context

Phase 3 checklist (policy)

- ☐ Default-deny posture defined for sensitive scope
- ☐ ADM-imported rules refined; Simulation run ≥ 1 week
- ☐ Change tickets for false-positive fixes; exception register updated
- ☐ **Export:** policy export + simulation report

Phase 4 checklist (operate)

- ☐ Enforcement enabled on pilot scope; negative test recorded in Denied Connections
- ☐ Quarterly pack: inventory, policy, denies, vulns, audit log (see playbook)
- ☐ SIEM integration verified (sample events)
- ☐ **Export:** enforcement screenshot + quarterly binder

What CSW evidence does not replace

Physical access, HR/training records, encryption key management, signed BAAs/vendor contracts, formal pen tests, and assessor attestation still require separate programmes. CSW addresses the **workload-resident** slice: segmentation, flows, process context, vuln reachability, and change drift.

CSW effectiveness for this framework

- Zones & conduits segmentation on IT-side OT boundaries
- SR control evidence; pair with Cyber Vision/Claroty for device layer
- Jump host and historian path visibility

Compared to manual programmes: static diagrams and annual firewall samples age immediately; CSW ties evidence to **live workload behaviour** and produces queryable exports on demand — supporting "operating effectively" language in PCI v4.0, SOC 2 CC7, and HIPAA risk analysis.

1. Overview

IEC 62443 provides a lifecycle-oriented security framework for industrial automation and control systems. **IEC 62443-3-3** specifies foundational requirements (FRs) and associated **system requirements (SRs)** for IACS components and systems. **IEC 62443-2-1** defines requirements for an IACS **security management system** operated by the asset owner. CSW contributes **technical evidence** for restricted data flow, access enforcement, integrity-related visibility, detection support, and availability-oriented anomaly awareness on the **IT workloads** that border or support IACS — not as a substitute for OT protocol inspection or safety-instrumented logic.

IEC 62443 structure mapped to CSW capabilities

IEC 62443 topic	Typical artifacts	Relevant CSW capabilities
FR 5 / SR 5.1–5.4 — Restricted data flow (zones & conduits)	Zone diagrams, conduit allowlists, firewall/ACL rules	Microsegmentation workspaces; scope hierarchy mirroring zones; default-deny with explicit conduits

IEC 62443 topic	Typical artifacts	Relevant CSW capabilities
FR 1 / SR 1.1–1.13 — Identification, authentication, access control	Account lifecycle, RBAC, remote access paths	Identity-aware policies (where integrated); least-privilege allowlists; jump-host-only admin paths
FR 3 / SR 3.1–3.9 — System & data integrity	Software inventory, patch posture, change detection	Process/binary inventory; listening-port baseline; vulnerability context; drift alerts
FR 6 / SR 6.1–6.2 — Event monitoring & timely response	SOC reference designs, timelines, ticket exports	Flow + process forensics; alert correlation inputs to SIEM
FR 7 / SR 7.1–7.8 — Resource availability	SLAs, capacity plans, DoS playbooks	Flow anomaly detection (connection/session volume spikes); early warning on saturation patterns
62443-2-1 — Security management / operations	KPIs, dashboards, management review	Continuous inventory/policy dashboards; scheduled evidence export

Regional or contractual note: Many asset owners implement 62443 requirements through customer purchase specifications or national interpretations. Map this reference design’s **SR** references to *your* contractual clauses and audit questionnaires; numbering alone is not sufficient for certification evidence without your target SL and scope.

2. Pre-Deployment Checklist

Before deploying CSW sensors on IACS-adjacent IT estates, confirm:

- ☐ CSW cluster (SaaS or on-prem) is provisioned; **air-gap / data residency** rules reviewed if telemetry leaves the plant DMZ
- ☐ Network path from workloads to CSW cluster (typically **443 outbound**); proxy exceptions documented
- ☐ Linux/Windows agent compatibility verified for **engineering workstations, jump servers, historians, AD/PKI**, and cloud broker VMs
- ☐ Cloud connectors configured where plant data lands in **AWS / Azure / GCP** (historian mirrors, IoT hubs, analytics lakes)
- ☐ Stakeholders engaged: **OT security lead, plant IT, integrator, SOC, and SMS owner** (62443-2-1)

- ❑ **Maintenance window** approved; **OT change freeze** rules respected — start in **Monitoring Only**
 - ❑ **PAS / OT tool** (Cyber Vision, Claroty, Nozomi) inventory export available for cross-walk of hostnames / VLANs / cell IDs
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3. Phase 1 — Sensor Deployment (Days 1–5)

3.1 On-premises IT adjacent to IACS

Install software sensors on representative tiers: EWS images, jump hosts, historian/MES application servers, AD connectors, vendor VPN concentrators — *never* on safety PLCs or real-time controllers if those run unsupported or prohibited agent OSes.

```
# Linux (RHEL/CentOS/Ubuntu family)
sudo rpm -ivh tet-sensor-<version>.rpm      # RHEL/CentOS-compatible
sudo dpkg -i tet-sensor-<version>.deb       # Debian/Ubuntu
```

```
# Verify sensor service
sudo systemctl status csw-agent
sudo systemctl enable --now csw-agent
```

Windows (engineering workstation build):

```
# Example: install from staged MSI (version/path per release)
msiexec /i "C:\Deploy\tet-sensor-<version>.msi" /qn
Get-Service | Where-Object { $_.Name -like "*tet*" -or $_.Name -like
"*csw*" }
```

Initial posture:

- Enforcement: **Monitoring Only** (no blocks until ADM + simulation complete)
- Collection: **process hash, network flow, listening services, vulnerability exposure** (where licensed)
- Tags from day one: zone:<cell>, conduit:<id>, iqs:iacs-adj, sl-target:SL2 (example)

3.2 Cloud & hybrid IACS data paths

Use **cloud connectors** when historian/analytics or IT control plane lives in public cloud:

```
CSW UI → Platform → External Orchestrators
→ Add AWS / Azure / GCP connector
→ Least-privilege IAM / service principal
→ Enable flow visibility (VPC/VNet flow ingestion where supported)
```

3.3 Sensor validation

```
CSW UI → Manage → Agents
→ Status: Active for all pilot hosts
→ Confirm telemetry (flow/process) within 15–30 minutes of install
```

→ Record agent build in baseline export (62443-3 SR 3 / change mgmt input)

4. Phase 2 — Scope & Inventory Design — Zones, Conduits, SL (Days 6–12)

4.1 Map CSW scopes to IEC zones & conduits

Model scopes to mirror **62443-3-2** zone definitions without renaming your-certified drawings — use **labels** that reference drawing IDs.

```
Root: Industrial-Enterprise
├─ Plant-<Site>-OT-Support          # IT that touches OT (EWS, jump,
historian app tier)
│   └─ Cell-A-Conduit-IT-OT        # Scoped to hosts in conduit
endpoints only
│       └─ Cell-B-Conduit-IT-OT
│           └─ Shared-Services-IACS # AD, PKI, patch, backup –
constrained to conduits
├─ Plant-DMZ-Brokers                # Data diodes / OPC brokers / MQTT
gateways (VM-based)
├─ Corporate-IT                     # Default no direct path to OT-
support
│   └─ Vendor-Remote-Access         # Termination servers for
integrator VPN
```

Conduit rule of thumb: One CSW policy workspace per **documented conduit** (or group of symmetric conduits) so evidence maps 1:1 to the ZCRD.

4.2 Discovery filters for IACS-adjacent candidates

Filter: Historian-MES-Tier

- Process name contains: historian, mes, opc*, scada, "wonderware"
- Ports: 1433, 1521, 5432, 135, 502 (context-dependent – validate in YOUR env)
- Tag pas:ot-adjacent (if synchronized from PAS)

Filter: Jump-EWS

- Hostname regex: (?i)(ews|eng|hmi-jump|vendor-jump)
- User workload tag: role=engineering

Validate every automated filter with **OT SME sign-off** — **SR 5** evidence must match the *as-engineered* conduit list, not a guessed list.

4.3 Label strategy (SMS-friendly)

Label key	Examples	Purpose
zone	cell-a, line-3	Aligns to 62443 zone ID
conduit	c-ot-to-mes-01	Audit trail for restricted

Label key	Examples	Purpose
sl-target	SL2, SL3	data flow Drive rule strictness / monitoring cadence
ics-role	ews, jump, historian, broker	Policy grouping
owner	ot-sec, plant-it	Accountability in exports

5. Phase 3 — Application Dependency Mapping & Baseline (Days 13–28)

5.1 ADM workspace

ADM establishes **observed** vs **designed** conduits — essential for **SR 5** and access **SR 1** reviews.

CSW UI → Investigate → Application Dependency Mapping

- New Workspace
- Name: IACS-Conduit-Baseline-<Site>
- Scope: Plant-<Site>-OT-Support + DMZ brokers
- Window: minimum 2–4 weeks (capture patch Tuesday, recipe changes, campaigns)
- Enable process context; retain ≥ 30 days if storage policy allows

5.2 ADM review checklist (62443-oriented)

Investigation question	SR / FR linkage
Which corporate subnets talk to historians or OPC brokers?	SR 5 — restricted data flow
Are there HTTPS/LD/RDP paths bypassing EWS/jump?	SR 1 — access paths
Do EWS images reach the internet directly?	SR 5, SR 1
New binary or service on jump host?	SR 3 — integrity
Sudden east-west fan-out from a broker VM?	SR 6 — detection; SR 7 — availability stress

5.3 Baselining for integrity (SR 3 inputs)

Export **inventory snapshots** after accepted build state:

CSW UI → Investigate → Inventory (or equivalent workload inventory view)

- Filter: scope = Plant-<Site>-OT-Support
- Export: processes, packages, listening ports, hashes (as

available)

→ Store in secure evidence library with version + timestamp

Repeat on **known-good** patch cycles to update baseline registers.

6. Phase 4 — Policy Design & Enforcement (Days 29–45)

6.1 Conduit-aligned policy framework

Translate **allowlisted conduits** into CSW rules. **Example pattern** — adjust ports, scopes, and identities to your ZCRD.

Absolute denies (illustrative):

DENY: Corporate-IT → Plant-<Site>-OT-Support (all)

EXCEPTION: documented Intermediary / jump subnet only

DENY: Vendor-Remote-Access → any except Plant-DMZ-Brokers & Jump hosts

DENY: Plant-<Site>-OT-Support → Internet (0.0.0.0/0)

EXCEPTION: explicit patch/CDN allowlist + CSW SaaS egress if required

Conduit allows (illustrative):

ALLOW: Plant-DMZ-Brokers ↔ Historian-MES-Tier

- TCP 135, 49320–49330 (DCOM range – validate), OPC-UA 4840 (if used)

ALLOW: Jump-EWS → Plant-<Site>-OT-Support

- RDP 3389 / SSH 22 only from Jump-EWS scope

ALLOW: Plant-<Site>-OT-Support → AD-PKI

- LDAPS 636, Kerberos 88, DNS 53 – block cleartext LDAP 389 where possible

Audit unmatched:

LOG + ALERT: any flow not matching explicit conduit rules

6.2 Policy workspace

CSW UI → Defend → Segmentation

→ New Workspace: IEC62443-Conduits-<Site>

→ Import ADM-suggested rules → reconcile with ZCRD

→ Mode: Simulation (minimum 2 weeks typical) → Enforcement by conduit tier

6.3 Enforcement progression

Step	Mode	Objective
1	Simulation	Prove OTSupport apps still function; tune OPC/RPC ranges
2	Selective enforcement	Lock down highest-risk corporate lateral paths first
3	Full enforcement on IT border	Default deny except conduits; maintain break-glass procedure

Integrator coordination: Provide **simulation reports** before enabling blocks on lines under warranty support.

7. Phase 5 — Monitoring, Availability Signals & SMS Feeds

7.1 Alerts aligned to 62443 operations

Alert	Example trigger	SR mapping
Undocumented conduit attempt	New src → dst across OT-support boundary	SR 5
Cleartext admin	LDAP/HTTP to AD from EWS	SR 1, SR 3
New listening service on jump	Port opens not in baseline	SR 3
Flow anomaly / storm	Connection rate >> baseline	SR 6 (detect), SR 7 (availability)
Critical CVE on historian tier	CVSS ≥ threshold	SR 3, 62443-2-1 risk treatment
Sensor offline	Loss of heartbeats	SR 6 — visibility gap

7.2 IEC 62443-2-1 continuous monitoring dashboard

Schedule **weekly** management-visible tiles (exact UI path may vary by release):

- **Scope coverage** — % of IACS-adjacent IT with active agent
- **Policy drift** — rules changed since last approval export
- **Open simulation violations** — unreviewed would-block flows
- **Top talkers** across conduits — compare to PAS asset roles

Export snapshots into your SMS review record **monthly** minimum.

8. Control-by-control mapping — Framework requirement → CSW → evidence

Use this table to **start** questionnaire mapping. Official conformity assessment requires your **SL-T**, **scope statement**, and **test methods** from the certification body or customer spec.

8.1 SR 5 — Restricted data flow (zones & conduits)

IEC 62443-3-3

requirement (summary)	CSW capability	Evidence produced
SR 5.1 — Segmentation / zones	Scope hierarchy; segmentation workspaces; labels zone, conduit	Scope export; diagram cross-walk memo
SR 5.2 — Segmentation for zones of differing security requirements	Tiered scopes by SL labels; stricter workspaces for higher SL	Policy diff between SL scopes
SR 5.3 — Conduit control	Allowlist rules per conduit; default deny	Policy JSON/export + simulation/enforcement logs
SR 5.4 — Covert channel mitigation (as applicable at IT layer)	Egress restrictions; anomaly detection on volume	Flow anomaly reports + change tickets

8.2 SR 1 — Identification, authentication, and access control

IEC 62443-3-3

requirement (summary)	CSW capability	Evidence produced
SR 1.1–1.13 (human / process / service access paths)	Identity-aware policies (when integrated); admin path lock-down; least privilege allowlists	Flow logs with user/process context; policy allow/deny history
Remote access paths via jump	Rules limiting src to jump scope; deny corporate → OTSupport direct	ADM + enforcement hit logs

8.3 SR 3 — System integrity

IEC 62443-3-3

requirement (summary)	CSW capability	Evidence produced
SR 3.1–3.9 (software integrity, malware deterrence inputs, etc.)	Process & binary inventory; package visibility; vulnerability data	Inventory export; drift alerts; VA connectors / built-in reports

IEC 62443-3-3 requirement (summary)	CSW capability	Evidence produced
Unauthorized software indicators	New process / listener alerts vs baseline	Ticket + before/after inventory

8.4 SR 6 — Event monitoring & timely response

IEC 62443-3-3 requirement (summary)	CSW capability	Evidence produced
SR 6.1 — Audit logging support	Flow + process retention; export APIs / UI exports	Raw & summary logs with UTC stamps
SR 6.2 — Continuous monitoring inputs	Alerts; SIEM forwarders	Alert rule catalog; SOC ingest proof

8.5 SR 7 — Resource availability

IEC 62443-3-3 requirement (summary)	CSW capability	Evidence produced
SR 7.1–7.8 (DoS considerations at system level)	Flow volumetrics; session rate anomalies; saturation precursors	Time-series anomaly exports + IR narrative

8.6 IEC 62443-2-1 — Security management system (technical contributors only)

62443-2-1 theme (summary)	CSW capability	Evidence produced
Operations monitoring & KPIs	Dashboards; scheduled exports	Monthly PDF/CSV appendices to SMS review
Incident response support	Forensic drill exports	Tabletop package — flow + process trace

9. Vulnerability, patch & compensating controls

9.1 Vulnerability visibility

CSW UI → Investigate → Vulnerability Report
 → Scope: Plant-<Site>-OT-Support
 → Filter: CVSS ≥ 7.0 (example)
 → Export CSV for SMS risk register linkage

9.2 Compensating controls when patching waits for outage

- Narrow allow rules to **known peer scopes** only
- Raise **alert severity** on vulnerable process names
- Capture **full flow logs** to affected workload until patched

10. Forensics & incident reconstruction

CSW UI → Investigate → Flow Search

- Time range: incident window (extend for long-running OT campaigns)
- Source/Destination: suspected broker or EWS
- Export: CSV/JSON with process + user context

CSW UI → Investigate → Process Search

- Parent/child chain for suspicious binaries on jump hosts

Retain exports per **your** SMS / legal hold policy; correlate with **PAS** PCAP or ICS alerts for full OT picture.

11. Audit preparation & evidence export

11.1 Quarterly evidence pack (adjust to customer cadence)

Evidence item	CSW source	Typical IEC mapping
Conduit policy export	Defend → workspace export	SR 5.x
Simulation vs enforcement history	Policy lifecycle / audit log	SR 5, SR 1
ADM cluster report	ADM workspace	SR 5, SR 1
Inventory baseline delta	Inventory exports	SR 3.x
Vulnerability exposure	Vulnerability report	SR 3, 62443-2-1
Alert & flow excerpts	Flow Search, Alerts	SR 6 , incident packs
Coverage report	Agent inventory	62443-2-1 monitoring scope

11.2 Bash helper examples (on analysis workstation)

After UI export to /evidence/csw/:

```
# Verify exported policy archive integrity (after UI download)
shasum -a 256 /evidence/csw/iec62443-policy-export-YYYYMMDD.zip |
tee /evidence/csw/SHA256SUMS
```

```
# Redact non-production subnets before sharing with integrator
perl -pe 's/\b10\.\d{1,3}\.\d{1,3}\.\d{1,3}\b/10.REDACTED.0.0/g' \
/evidence/csw/flows-incident-window.csv > /evidence/csw/flows-incident-window-redacted.csv
```

12. Boundaries — what CSW does **not** cover

- **Level 0–2 OT devices** (PLCs, IEDs, devices without supported agents) — use PAS/OT security tools; CSW does not inspect proprietary fieldbus payloads at wire speed.
 - **Safety instrumented functions** — CSW is not a SIL-rated safety device; engineering lifecycle evidence stays with SIS vendor tools.
 - **Physical security & personnel screening** — access badges, gates, contractor vetting — outside CSW.
 - **Cryptographic key management policy** — CSW may enforce **paths** (e.g., TLS-only) but does not replace PKI governance or HSM decisions.
 - **62443-2-1 organizational processes** — CSW supplies **telemetry and policy exports**; it does not author your procedures, training roster, or supplier security clauses.
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13. Common pitfalls

Pitfall	Mitigation
Mis-mapping CSW scopes to electrical zones instead of cyber zones	Anchor labels to approved cyber ZCRD IDs
Blocking dynamic RPC/OPC ranges without ADM	Long simulation + integrator test scripts
Deploying only on corporate IT	OT-adjacent IT is the compliance-critical tier
Ignoring cloud shadow IT paths to plant data	Enable cloud connectors for mirrored historians
Expecting CSW to replace OT IDS	Maintain Cyber Vision / Claroty / Nozomi for ICS context

Related Frameworks

- [NERC CIP](#) — analogous IT/OT boundary evidence pattern for BES entities.
 - [NIST SP 800-82](#) — operational technology companion guidance (external).
 - [ISO/IEC 27001:2022](#) — when plants pair 62443 with enterprise ISMS audits.
 - [NIST SP 800-207](#) — zero trust patterns for conduit enforcement narratives.
 - [UK NCSC CAF v3.2](#) — zones and conduits are the reuse path for CAF B5 on the IT side of an IT/OT boundary.
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Appendix A — Sample segmentation policy fragment (illustrative YAML-style)

Do not paste verbatim into production. Replace scopes, addresses, and L4 tuples with values from your ZCRD. Syntax mirrors conceptual CSW rule metadata; exact API/UI format follows product version.

```
workspace: IEC62443-Conduits-Site01
mode: simulation           # change to enforcement after sign-off
default_action: deny
rules:
  - name: conduit-opcua-historian
    action: allow
    src_scope: Plant-DMZ-Brokers
    dst_scope: Historian-MES-Tier
    services:
      - tcp/4840
  - name: admin-via-jump-only
    action: allow
    src_scope: Jump-EWS
    dst_scope: Plant-Site01-OT-Support
    services:
      - tcp/22
      - tcp/3389
  - name: log-unknown-east-west
    action: alert
    src_scope: Plant-Site01-OT-Support
    dst_scope: Plant-Site01-OT-Support
    match: not_in_conduit_allowlist
```

Appendix B — PAS correlation workflow

1. Export **asset list** from Cyber Vision / Claroty / Nozomi (IP, MAC, VLAN, firmware).
 2. Join on **IP or hostname** with CSW inventory.
 3. Flag **CSW-only** or **PAS-only** assets — resolve before **SR 5** attestation.
 4. For discrepancies, **field-verify** with plant engineering.
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Document prepared for industrial account engagements. Replace site names, drawing references, and SL targets with customer-specific values before delivery.